Université de Genève

Imagerie Numérique
13X004

TP 10: 1D Discrete Fourier Transform

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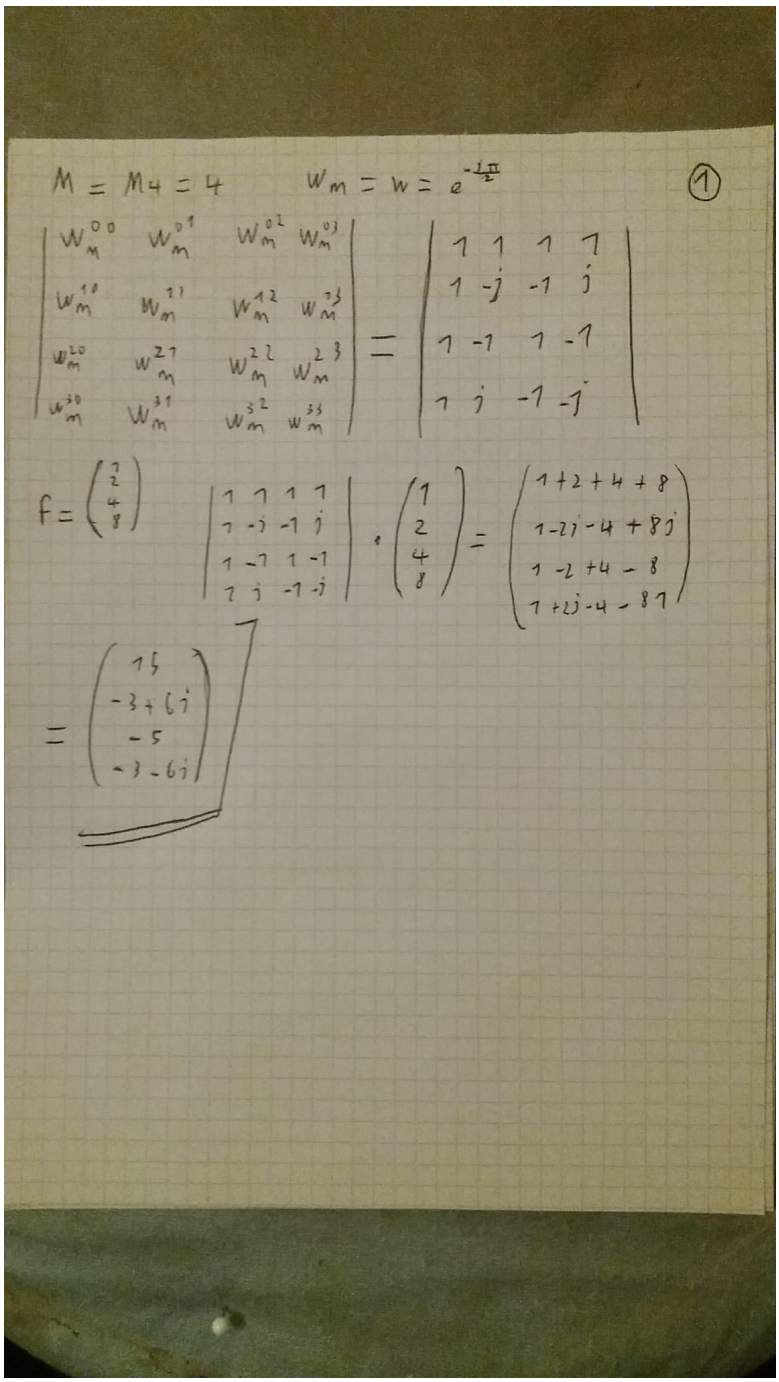
Imagerie numérique TP10

Exercice 1 DFT Exploration (4 points)

(a) We consider the following discrete signal : f = [1,2,4,8]

I computed by hand the DFT of f:



$M = M_4 = 4 \qquad W_m = W = e^{-\frac{j\pi}{2}}$

1

$$\begin{vmatrix} W_m^{00} & W_m^{01} & W_m^{02} & W_m^{03} \\ W_m^{10} & W_m^{11} & W_m^{12} & W_m^{13} \\ W_m^{20} & W_m^{21} & W_m^{22} & W_m^{23} \\ W_m^{30} & W_m^{31} & W_m^{32} & W_m^{33} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{vmatrix}$$

$$f = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix} \qquad \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{vmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \end{pmatrix} = \begin{pmatrix} 1+2+4+8 \\ 1-2j-4+8j \\ 1-2+4-8 \\ 1+2j-4-8j \end{pmatrix}$$

$$= \begin{pmatrix} 15 \\ -3+6j \\ -5 \\ -3-6j \end{pmatrix}$$

Figure 1: 00.jpeg

In python (ex1.py function a()) I found [15.+0.j -3.+6.j -5.+0.j -3.-6.j], that is the same answer.
We have a non zero imaginary part beacause the signal is not symmetric.
For instance a function with the form [a,b,a,b] (a periodic one) we can get:

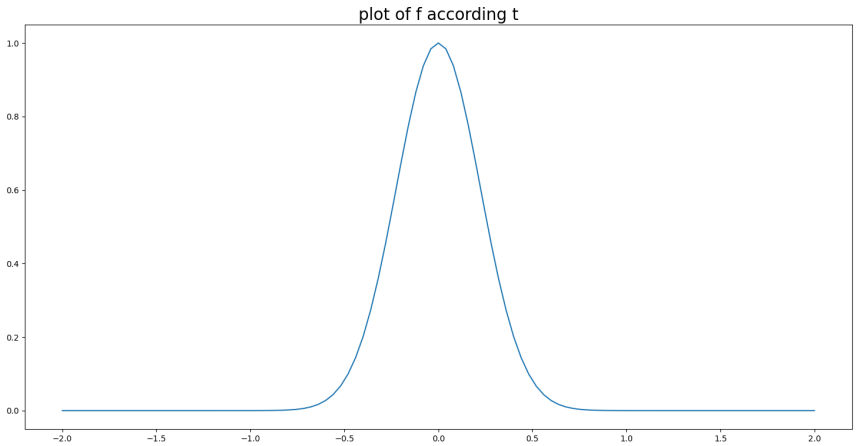
$$g = \begin{pmatrix} a \\ b \\ b \\ a \end{pmatrix} \quad \text{so} \quad \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{vmatrix} \cdot \begin{pmatrix} a \\ b \\ b \\ a \end{pmatrix} = \begin{pmatrix} a+b+a+b \\ a-bj-a+bj \\ a-b+a-b \\ a+bj-a-bj \end{pmatrix}$$

$$= \begin{pmatrix} 2a+2b \\ 0+0j \\ 2a-2b \\ 0+0j \end{pmatrix} = \boxed{\begin{pmatrix} 2(a+b) \\ 0 \\ 2(a-b) \\ 0 \end{pmatrix}}$$

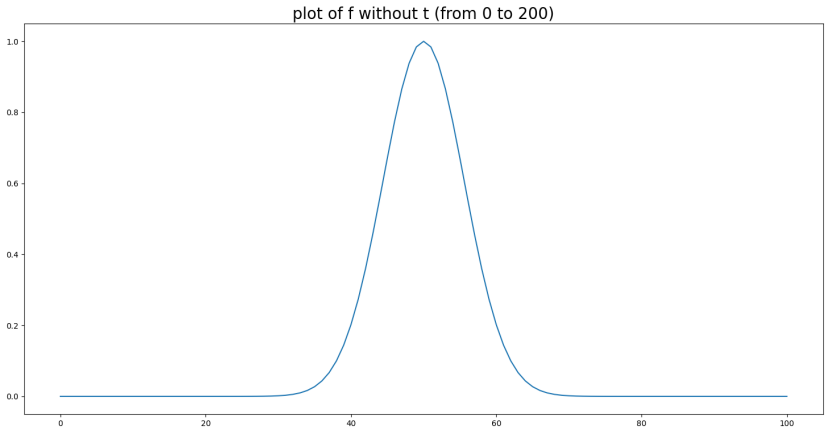
1

(b) We now turn to a more involved example.

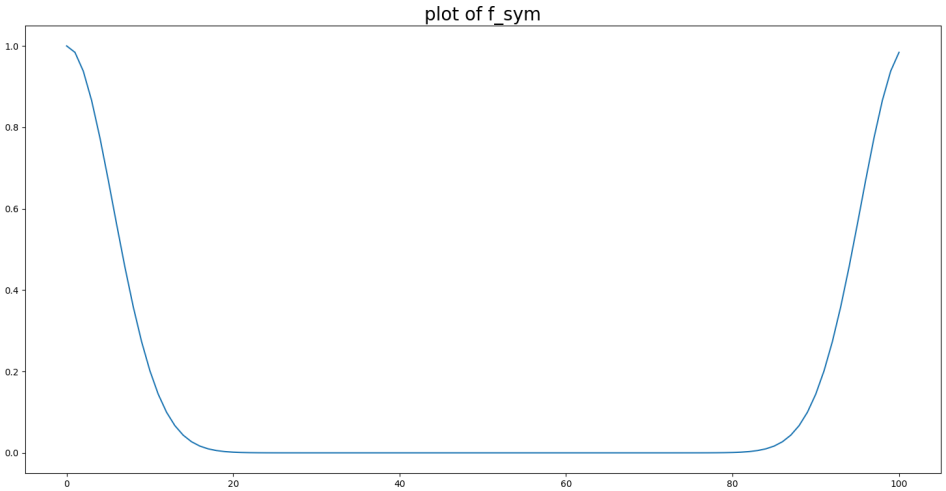
I defined a variable on the interval [-2,2] with 101 samples and a signal f= exp(-10*t^2)
I did a plot with plt.plot(t,f).

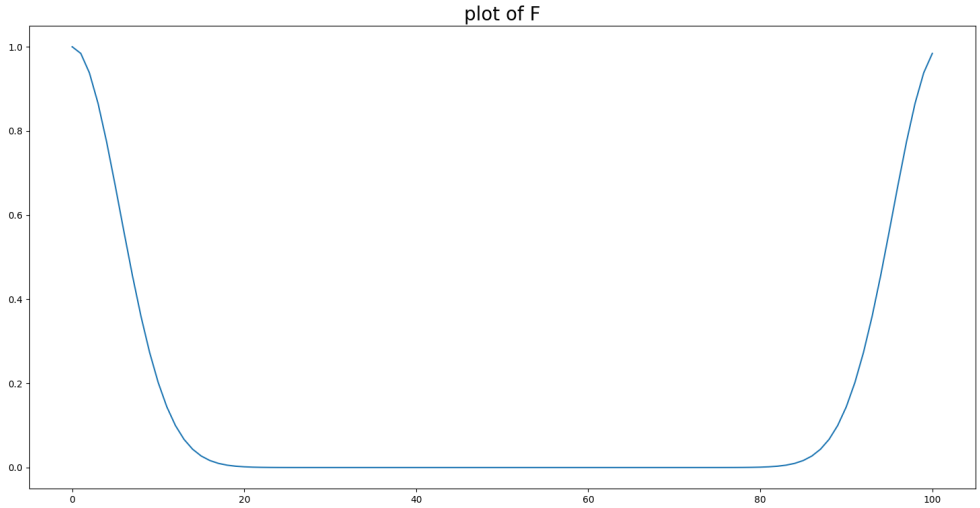


I also did a plot with plt.plot(f) (a shifted one).

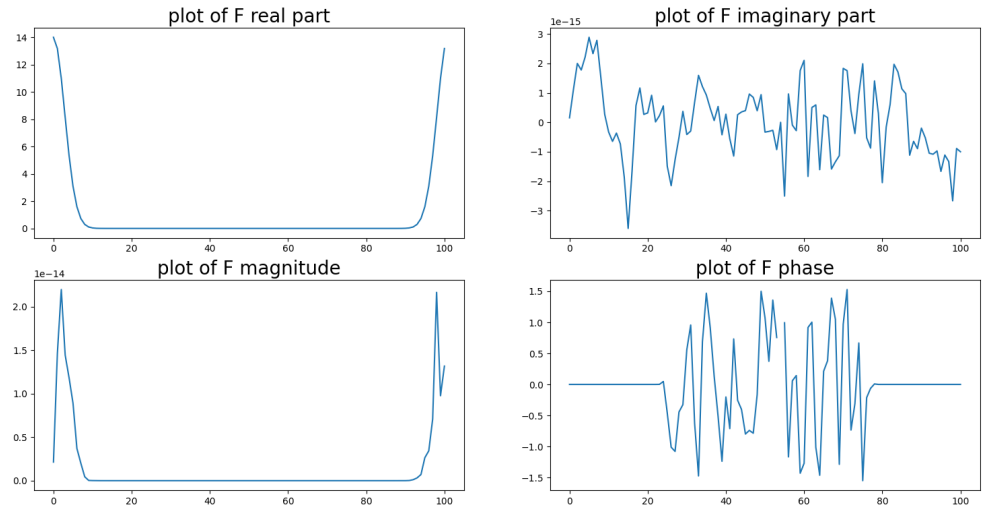


I defined a symmetric signal from f by rolling it: f_sym= np.roll(f,51) and ploted this new function. After that I computed F= np.fft.fft(f_sym) and ploted this new function.

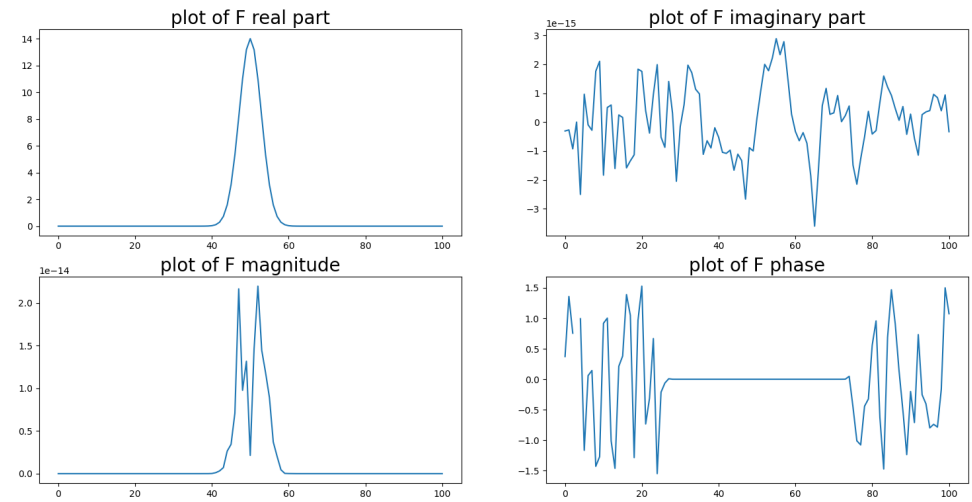




I computed D = np.fft.fft(f_sym) and represented it in four different plots.



To have a better result, I modify F using F= np.fft.fftshift(F).

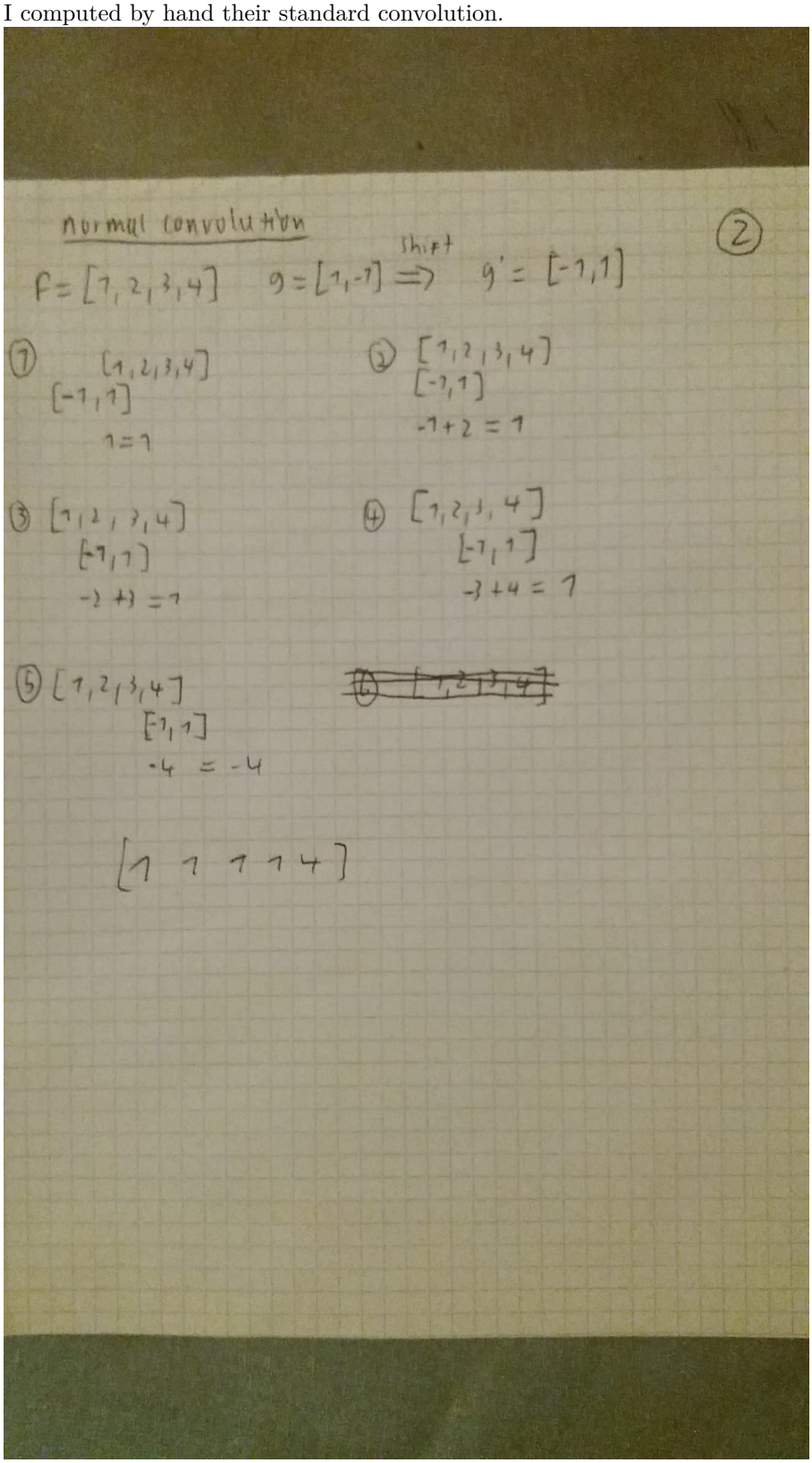


Exercice 2. Circular convolution

We consider the two signals f=[1,2,3,4] and g=[1.-1].

(a)

I computed by hand their standard convolution.



normal convolution

2

$f = [1, 2, 3, 4]$ $g = [1, -1] \xRightarrow{\text{shift}} g' = [-1, 1]$

1
[1, 2, 3, 4]
[-1, 1]
1 = 1

2
[1, 2, 3, 4]
[-1, 1]
-1 + 2 = 1

3
[1, 2, 3, 4]
[-1, 1]
-2 + 3 = 1

4
[1, 2, 3, 4]
[-1, 1]
-3 + 4 = 1

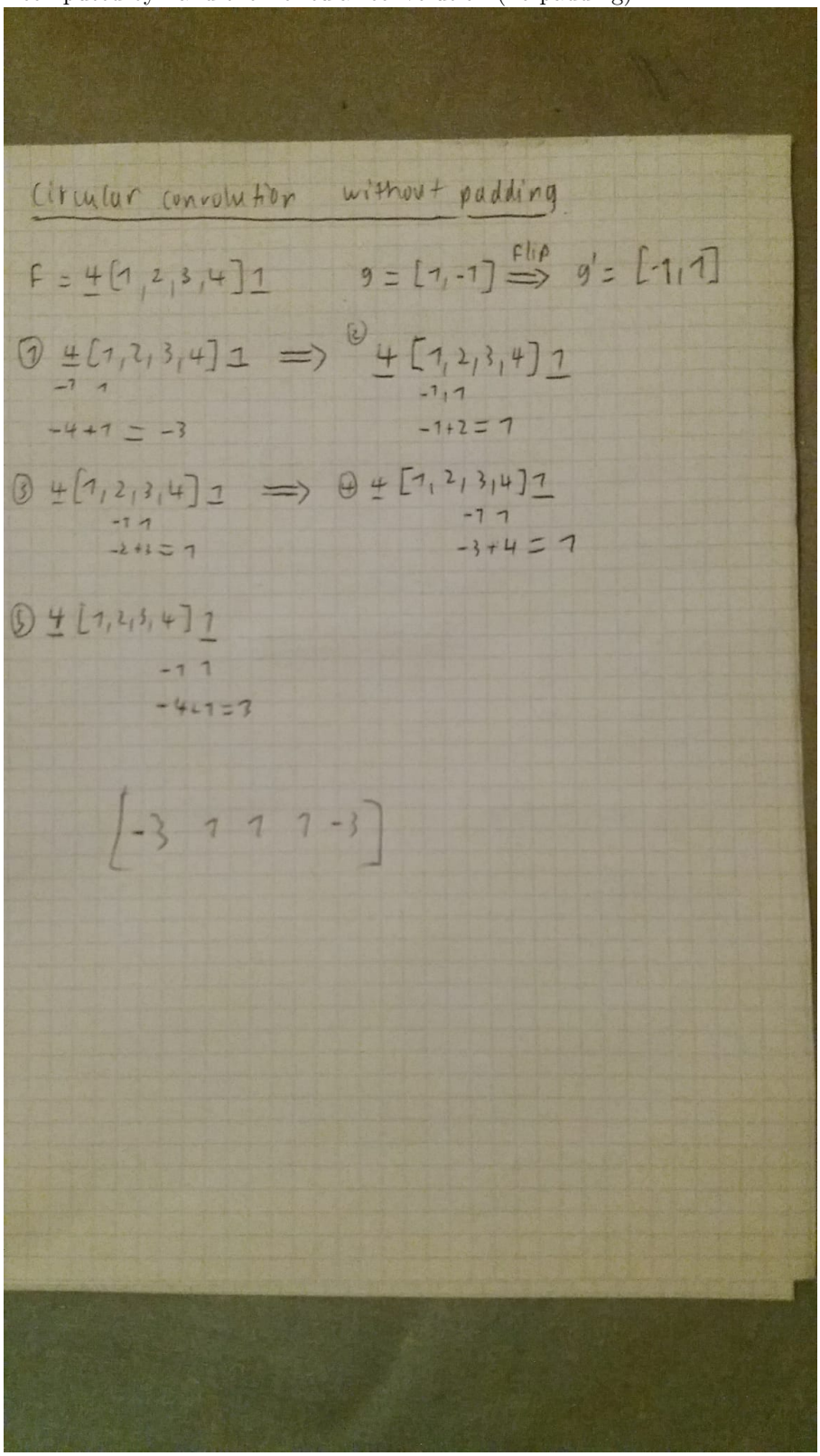
5
[1, 2, 3, 4]
[-1, 1]
-4 = -4

~~6 [1, 2, 3, 4]~~

[1 1 1 1 4]

(b)

I computed by hand their circular convolution (no padding)



Circular convolution without padding

$f = \underline{4}[1,2,3,4]\underline{1}$ $\quad g = [1,-1] \xRightarrow{\text{flip}} g' = [-1,1]$

① $\underline{4}[1,2,3,4]\underline{1} \Longrightarrow$ ② $\underline{4}[1,2,3,4]\underline{1}$

$-1 \quad 1$ $\qquad -1, 1$

$-4+1 = -3$ $\qquad -1+2 = 1$

③ $\underline{4}[1,2,3,4]\underline{1} \Longrightarrow$ ④ $\underline{4}[1,2,3,4]\underline{1}$

$-1 \quad 1$ $\qquad -1 \quad 1$

$-2+3 = 1$ $\qquad -3+4 = 1$

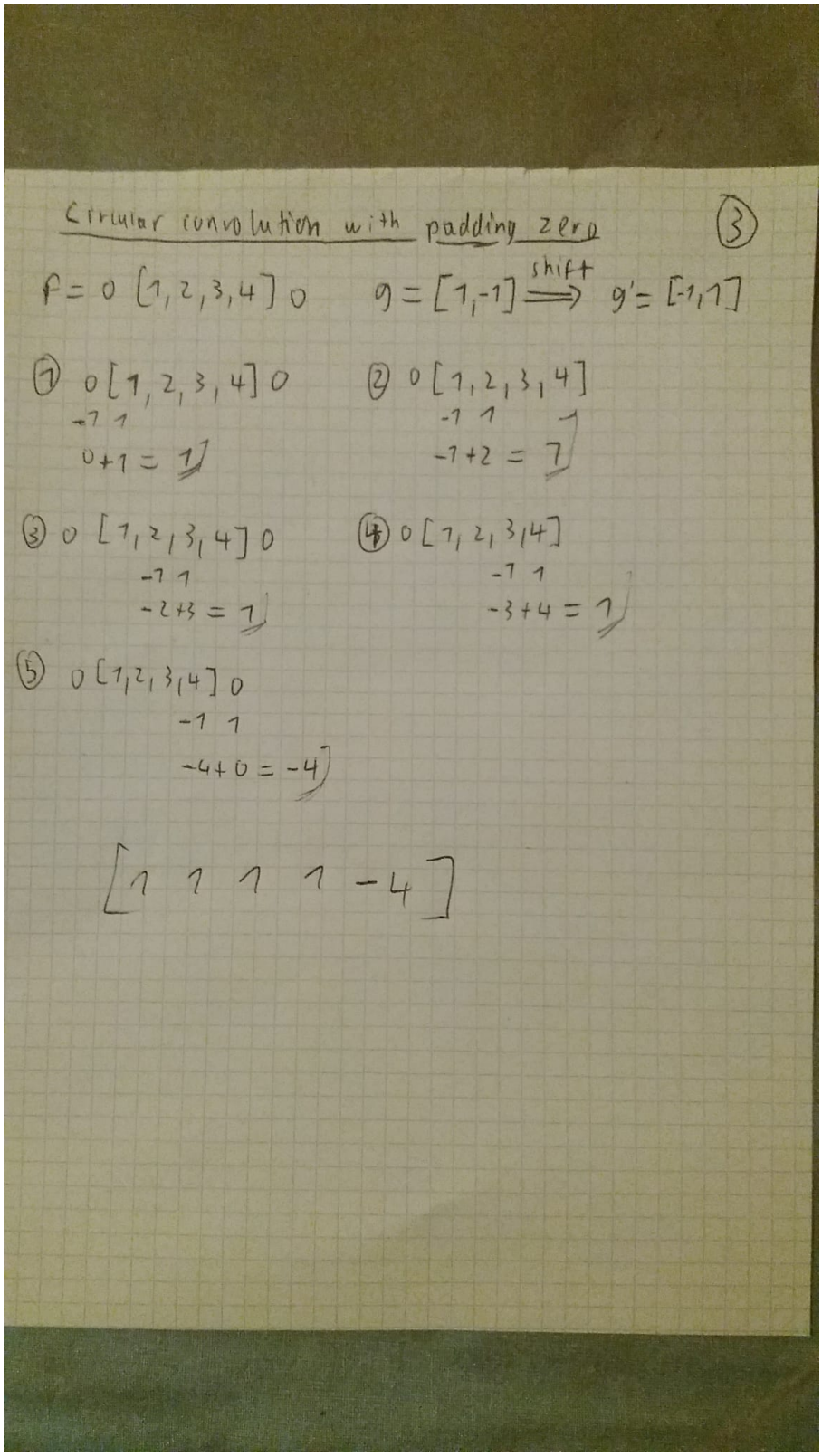
⑤ $\underline{4}[1,2,3,4]\underline{1}$

$-1 \quad 1$

$-4+1 = 3$

$[-3 \quad 1 \quad 1 \quad 1 \quad -3]$

I computed by hand their circular convolution (zero padding)



Circular convolution with padding zero (3)

$f = 0\ [1,2,3,4]\ 0 \qquad g = [1,-1] \xRightarrow{\text{shift}} g' = [-1,1]$

(1) $0\ [1,2,3,4]\ 0$
$-1\ 1$
$0+1 = 1$

(2) $0\ [1,2,3,4]$
$-1\ 1$
$-1+2 = 1$

(3) $0\ [1,2,3,4]\ 0$
$-1\ 1$
$-2+3 = 1$

(4) $0\ [1,2,3,4]$
$-1\ 1$
$-3+4 = 1$

(5) $0\ [1,2,3,4]\ 0$
$-1\ 1$
$-4+0 = -4$

$[1\ 1\ 1\ 1\ -4]$

With zero padding, I found the same result as a normal convolution.

With python I found respectively:
[1 1 1 1 -4]
[-3 1 1 1 -3]
[1 1 1 1 -4]

There are the expected values.